

Answers From Chatgpt

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1 Explanation of the $SU(2)$ Global Anomaly Remark

We explain the logic of the remark step by step.

1.1 $SU(2)$ Gauge Theory in Four Dimensions

The weak interaction is described by an $SU(2)$ gauge theory. Therefore, on our four-dimensional spacetime M there is an $SU(2)$ principal bundle

$$P \rightarrow M.$$

On a local patch we may treat spacetime as $\mathbb{R}^4$, and a gauge transformation is a map

$$g : \mathbb{R}^4 \rightarrow SU(2).$$

1.2 Compactification to S^4 and Large Gauge Transformations

Assume the gauge transformation becomes trivial at spatial infinity:

$$g(x) \rightarrow \mathbf{1} \quad \text{as } |x| \rightarrow \infty.$$

This allows us to compactify $\mathbb{R}^4$ by adding a point at infinity:

$$\mathbb{R}^4 \cup \{\infty\} \cong S^4,$$

with $g(\infty) = \mathbf{1}$. Hence gauge transformations are classified by maps

$$g : S^4 \rightarrow SU(2) \simeq S^3.$$

The homotopy classes of such maps are elements of

$$\pi_4(S^3) \simeq \mathbb{Z}_2 = \{0, 1\}.$$

The nontrivial class (1) corresponds to a “large” gauge transformation.

1.3 SU(2) Doublet Fermions

Fermions charged under the weak force transform in the fundamental (doublet) representation $\mathbf{2}$ of $SU(2)$. A single such fermion field is a section of the associated vector bundle

$$\mathbf{2} \otimes S \longrightarrow M,$$

where S is the spinor bundle. For our purposes, the only important fact is:

A single SU(2) doublet fermion is one section of a vector bundle.

1.4 Witten's SU(2) Global Anomaly

A key result of Witten (1982) states that under a large gauge transformation $g : S^4 \rightarrow S^3$, the partition function of a single $SU(2)$ doublet fermion transforms as

$$Z_{\text{single doublet}} \longrightarrow (-1)^s Z_{\text{single doublet}},$$

where $s = [g] \in \pi_4(S^3) \simeq \mathbb{Z}_2$. Thus

$$s = 0 : Z \text{ invariant}, \quad s = 1 : Z \mapsto -Z.$$

Therefore:

A single SU(2) doublet suffers from a global SU(2) anomaly.

1.5 Multiple Doublets and Anomaly Cancellation

If there are N doublets, the total transformation is

$$Z \longrightarrow (-1)^{Ns} Z.$$

The theory is gauge invariant for all gauge transformations if and only if N is even. In the Standard Model, one generation contains

$$3 \text{ quark doublets} + 1 \text{ lepton doublet} = 4 \text{ doublets}.$$

Hence

$$(-1)^{4s} = +1,$$

and the anomaly cancels.

This explains, for instance, why one cannot simply change $SU(3)$ color to $SU(4)$: it would change the number of weak doublets, potentially making N odd and producing an inconsistency.

1.6 Summary

- Large $SU(2)$ gauge transformations are classified by $\pi_4(S^3) \simeq \mathbb{Z}_2$.
- A single doublet fermion changes sign under the nontrivial transformation.
- A consistent theory must have an *even* number of doublets.
- The Standard Model satisfies this automatically ($N = 4$ per generation).

2 Fiber Choice in the Covering Map $\mathbb{R} \rightarrow S^1$

Consider the covering map

$$p : \mathbb{R} \longrightarrow S^1, \quad p(x) = e^{2\pi i x}.$$

For any point $z = e^{2\pi i \alpha} \in S^1$, the fiber is

$$p^{-1}(z) = \{x \in \mathbb{R} \mid e^{2\pi i x} = z\} = \alpha + \mathbb{Z},$$

a translated copy of $\mathbb{Z}$. Thus every fiber is homeomorphic to the discrete set $\mathbb{Z}$.

In the long exact sequence of a fibration, one chooses a basepoint $b_0 \in S^1$ and defines the fiber as $p^{-1}(b_0)$. Although $b_0 = 1 \in S^1$ (the identity element) is the conventional choice, any b_0 works because S^1 is path-connected and homotopy groups based at different points are naturally isomorphic.

3 Homotopy Groups of the Discrete Space $\mathbb{Z}$

Although $\mathbb{Z}$ is not a manifold, it is a valid topological space and its homotopy groups are well-defined.

3.1 All Continuous Maps $S^n \rightarrow \mathbb{Z}$ Are Constant ($n \geq 1$)

Since $\mathbb{Z}$ is discrete, each singleton $\{k\} \subset \mathbb{Z}$ is open. A continuous map from a connected space S^n to a discrete space must have connected image, hence the image consists of only one point. Thus any map

$$f : S^n \rightarrow \mathbb{Z}, \quad n \geq 1$$

is constant.

3.2 Computation of $\pi_n(\mathbb{Z})$

If we fix the basepoint $k_0 \in \mathbb{Z}$, the only based map $S^n \rightarrow \mathbb{Z}$ is the constant map $f(x) \equiv k_0$. Hence, for $n \geq 1$,

$$\pi_n(\mathbb{Z}, k_0) = 0.$$

For $n = 0$, the path components of $\mathbb{Z}$ are just its points, so

$$\pi_0(\mathbb{Z}) \cong \mathbb{Z}.$$

3.3 Summary

$$\boxed{\pi_0(\mathbb{Z}) = \mathbb{Z}, \quad \pi_{n \geq 1}(\mathbb{Z}) = 0.}$$

3.4 Application: Computing $\pi_n(S^1)$ Using the Covering Space

From the fibration

$$\mathbb{Z} \longrightarrow \mathbb{R} \longrightarrow S^1,$$

the long exact sequence gives

$$\pi_n(S^1) \cong \pi_{n-1}(\mathbb{Z}).$$

Using our calculation of $\pi_{n-1}(\mathbb{Z})$, we obtain the classical result:

$$\pi_n(S^1) = \begin{cases} \mathbb{Z}, & n = 1, \\ 0, & n \neq 1. \end{cases}$$

4 Understanding why $\partial : \mathbb{Z} \rightarrow \mathbb{Z}$ must be multiplication by 2

Consider the fibration

$$SO(2) \longrightarrow SO(3) \longrightarrow S^2.$$

The long exact sequence in homotopy (for $n = 3$ in the book's notation) contains the segment

$$\pi_2(SO(2)) \longrightarrow \pi_2(SO(3)) \longrightarrow \pi_2(S^2) \xrightarrow{\partial} \pi_1(SO(2)) \longrightarrow \pi_1(SO(3)) \longrightarrow \pi_1(S^2).$$

We substitute the known homotopy groups:

$$\pi_2(SO(2)) = 0, \quad \pi_2(S^2) = \mathbb{Z}, \quad \pi_1(SO(2)) = \mathbb{Z}, \quad \pi_1(SO(3)) = \mathbb{Z}_2, \quad \pi_1(S^2) = 0.$$

Thus the exact sequence becomes

$$0 \longrightarrow \pi_2(SO(3)) \longrightarrow \mathbb{Z} \xrightarrow{\partial} \mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}_2 \longrightarrow 0.$$

4.1 Step 1: Determining the form of ∂

Exactness at $\pi_1(SO(2)) = \mathbb{Z}$ means

$$\text{Im}(\partial) = \ker(\text{mod } 2).$$

Since

$$\ker(\text{mod } 2) = 2\mathbb{Z},$$

we obtain

$$\text{Im}(\partial) = 2\mathbb{Z}.$$

Any group homomorphism $\partial : \mathbb{Z} \rightarrow \mathbb{Z}$ must have the form

$$\partial(n) = kn, \quad k \in \mathbb{Z}.$$

Its image is $k\mathbb{Z}$. To satisfy $\text{Im}(\partial) = 2\mathbb{Z}$, the only possibility is

$$k = \pm 2.$$

Hence

$$\partial(n) = \pm 2n.$$

This is the meaning of the sentence in the book:

“This forces $\partial : \mathbb{Z} \rightarrow \mathbb{Z}$ to be a multiplication by 2.”

4.2 Step 2: Computing $\pi_2(SO(3))$

Exactness at $\pi_2(S^2) = \mathbb{Z}$ implies

$$\text{Im}(\pi_2(SO(3)) \rightarrow \mathbb{Z}) = \ker(\partial).$$

Since $\partial(n) = \pm 2n$, its kernel is

$$\ker(\partial) = \{0\}.$$

Thus

$$\text{Im}(\pi_2(SO(3)) \rightarrow \mathbb{Z}) = 0.$$

But the map $0 \rightarrow \pi_2(SO(3))$ at the beginning of the sequence is injective, so the only way $\pi_2(SO(3))$ can map to 0 inside $\mathbb{Z}$ is if

$$\pi_2(SO(3)) = 0.$$

So the conclusion is:

$$\boxed{\pi_2(SO(3)) = 0.}$$

5 Unitarity in 2-2 scattering

The partial-wave coefficients are defined by projecting the T -matrix onto two-particle states of definite total momentum and total angular momentum,

$$f_J(p^2) \propto \frac{\langle p, J, \vec{m} | \hat{T} | p, J, \vec{m} \rangle}{\langle p, J, \vec{m} | p, J, \vec{m} \rangle}. \quad (1)$$

Because $\hat{T}$ is invariant under translations and under $SO(1, d-1)$, the matrix element in this basis is diagonal in $(J, \vec{m})$, so the proportionality constant is independent of p^2 and of $\vec{m}$. The result therefore depends only on the Lorentz-invariant combination $p^2 = s$.

We start from the angular form of the elastic unitarity equation,

$$2 \Im T(s, t(z)) = \int_{-1}^1 dz' w(z') \int_{-1}^1 dz'' w(z'') \mathcal{P}_d(z, z', z'') T(s, t(z')) T^*(s, t(z'')), \quad (2)$$

where

$$w(z) = (1 - z^2)^{\frac{d-4}{2}}.$$

Next we insert the partial-wave expansion,

$$T(s, t(z)) = \sum_{L=0}^{\infty} n_L^{(d)} f_L(s) P_L^{(d)}(z), \quad (3)$$

and substitute this expression into (1). At this stage we obtain

$$\begin{aligned} 2 \Im T(s, t(z)) &= \sum_{L, L'} n_L^{(d)} n_{L'}^{(d)} f_L(s) f_{L'}^*(s) \int_{-1}^1 dz' w(z') \mathcal{P}_d(z, z', z'') P_L^{(d)}(z') \\ &\quad \times \int_{-1}^1 dz'' w(z'') P_{L'}^{(d)}(z''). \end{aligned} \quad (4)$$

To isolate the coefficient $f_J(s)$ we project both sides of (3) onto the J -th partial wave, multiplying by

$$\frac{\mathcal{N}_d}{2} w(z) P_J^{(d)}(z)$$

and integrating over z . Using the projection formula

$$f_J(s) = \frac{\mathcal{N}_d}{2} \int_{-1}^1 dz w(z) P_J^{(d)}(z) T(s, t(z)), \quad (5)$$

the left-hand side becomes simply

$$\frac{\mathcal{N}_d}{2} \int_{-1}^1 dz w(z) P_J^{(d)}(z) 2 \Im T(s, t(z)) = 2 \Im f_J(s).$$

Now we insert the kernel decomposition (eq. (67))

$$\mathcal{P}_d(z, z', z'') = \frac{1}{2} \mathcal{N}_d^2 w(z') w(z'') \sum_{L=0}^{\infty} n_L^{(d)} P_L^{(d)}(z) P_L^{(d)}(z') P_L^{(d)}(z''), \quad (6)$$

and use the orthogonality relation

$$\frac{1}{2} \int_{-1}^1 dz w(z) P_J^{(d)}(z) P_L^{(d)}(z) = \frac{\delta_{JL}}{\mathcal{N}_d n_J^{(d)}}. \quad (7)$$

Substituting (5) into (3) and performing the two z', z'' integrals, only the term $L = L' = J$ survives due to (6), giving

$$\frac{\mathcal{N}_d}{2} \int dz w(z) P_J^{(d)}(z) (\text{RHS}) = \frac{(s - 4m^2)^{\frac{d-3}{2}}}{\sqrt{s}} |f_J(s)|^2.$$

Equating both sides yields the elastic partial-wave unitarity condition

$$2 \Im f_J(s) = \frac{(s - 4m^2)^{\frac{d-3}{2}}}{\sqrt{s}} |f_J(s)|^2, \tag{8}$$

which is eq. (68) of the main text.